

Day 8 – Shell Scripting: Arrays & Debugging

Objective: Learn arrays and debugging in bash scripts.

Arrays in Bash allow storing multiple values in one variable.

Example: `SERVERS=(web1 web2 db1)`

Access element: `${SERVERS[0]}`

Print all elements: `${SERVERS[@]}`

Loop through array:

```
for server in ${SERVERS[@]}; do echo Checking $server; done
```

Example server connectivity script:

```
SERVERS=(google.com github.com gitlab.com)
```

```
for server in ${SERVERS[@]}; do
```

```
ping -c 1 $server > /dev/null 2>&1
```

```
if [ $? -eq 0 ]; then echo "$server reachable"; else echo "$server unreachable"; fi
```

```
done
```

Debugging scripts:

```
set -x (enable debug)
```

```
set +x (disable debug)
```

Example debug script:

```
#!/bin/bash
```

```
set -x
```

```
date
```

```
whoami
```

```
set +x
```

Best Practices:

- Quote variables
- Use arrays for multiple servers
- Use debugging when troubleshooting scripts

Day 8 completed.